

Sachcith G N

B.Tech CSE (AI), Amrita Vishwa Vidyapeetham, Coimbatore
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Education

Amrita Vishwa Vidyapeetham, Coimbatore B.Tech CSE (AI), CGPA: **9.23/10.0**

Expected 2028

Technical Skills

Languages: Python, C++, C, MATLAB, Java, SQL

Systems & Software: Data Structures & Algorithms, Systems Programming, Embedded Systems, Concurrency, Database Systems, Operating Systems, Computer Networks, Object-Oriented Design, Distributed Systems, Multi-Threading

Machine Learning: Machine Learning, Computer Vision, Signal Processing

Frameworks & Tools: Linux, Git, Flask, Socket.IO, ESP32

Projects

ERP Management System

GitHub

- Architected and implemented a production ERP platform for a jewelry business using Flask, PostgreSQL, and Socket.IO, supporting inventory management, transaction processing, stock tracking, customer management, and authentication across 10+ relational database tables and validated against workloads exceeding **1,000** monthly transactions.
- Engineered a transaction-consistency framework that automates stock reconciliation, inventory updates, audit tracking, historical ledger correction, and business-rule validation while preserving data integrity across interconnected financial and inventory workflows.

Embedded Audio Intelligence System

GitHub

- Deployed a compressed MLP classifier by porting a Python training pipeline to Embedded C++, resolving watchdog-induced failures and optimizing feature extraction memory usage through custom buffering strategies that reduced auxiliary memory requirements by **50%**, enabling stable end-to-end inference on ESP32 across 100 benchmark runs.
- Built an end-to-end audio intelligence framework in C++ for ESP32, implementing on-device audio acquisition, DSP pipelines, STFT/DWT feature extraction, statistical descriptors, and neural-network inference without relying on high-level ML frameworks.

Othello AI Engine

GitHub

- Developed a complete Othello engine from scratch, implementing game-state management, legal move generation, board evaluation, and reversible state transitions.
- Implemented an Alpha-Beta Pruned Minimax search agent with custom heuristic evaluation, reducing search space while enabling competitive real-time gameplay through a Flask and Socket.IO web interface.

Publications

Manuscript Under Review – IEEE Transactions on Reliability

- Designed a YOLOv11-based photovoltaic fault detection framework achieving 98.1% mAP@0.5, and 6.4ms inference while outperforming 36+ prior approaches.

IEEE PES Hong Kong Conference Publication

Link

- Designed and evaluated a transformer-based load forecasting system integrating FEDFormer and cross-attention mechanisms, achieving RMSE 0.2243 and MAE 0.101.

Competitive Programming

- Member of Amrita Vishwa Vidyapeetham's ICPC team; active Codeforces participant with experience in algorithms, graph theory, dynamic programming, and contest problem solving.
- Solved 250+ problems across Codeforces and competitive programming platforms.